MuTau channel: event weights documentation

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Overview of event weights

All samples carry a per-event `weight` branch written by the processing code. The final weight stored in the output tree is a product of several factors:

$$w_{\text{final}} = w_{\text{norm}} \times w_{\text{muon}} \times w_{\text{rad}}$$

Factor	Symbol	Applied to
Sample normalization	w_{norm}	Signal, DY, ttbar
Muon scale factors	w_{muon}	Signal, DY, ttbar
Pixel radiation damage	w_{rad}	Signal only
QCD / Data	1	QCD, Data

Signal normalization weight (`sinal.py`)

Formula (per event, computed before any selection cut):

$$w_{\text{norm}} = w_{\text{SM}}[0] \times \frac{L \times \sigma_{\text{SM}}}{\sum w_{\text{SM}}[0]}$$

Quantity	Value	Source
L	$54900\ \text{pb}^{-1}$	2018 UL luminosity
σ_{SM}	0.0047 pb	SM reweighting weight 51
$\sum w_{\text{SM}}[0]$	402.661	Sum over all 299,979 events in the ntuple

Cross-check: $\sum w_{\text{norm}} = L \times \sigma_{\text{SM}} = 54900 \times 0.0047 = 258$ expected SM events before any cut. ✓

Code location: `sinal.py`, line 447:

```
weight_sample = weight_sm[0] * (54900.0 * 0.0047) / 402.661
```

Muon scale factors (sinal.py)

Applied **after** the muon+tau selection (step 2 in the cutflow):

$$w_{\text{muon}} = \text{SF}_{\text{trig}} \times \text{SF}_{\text{ID+ISO}} \times \text{SF}_{\text{reco}}$$

Factor	Source	η / p_T binned?
Trigger SF (mu_trig)	muon_neg18_lowTrig.txt, muon_pos18_lowTrig.txt, muon_18_highTrig.txt	Yes
ID+ISO SF (mu_idiso)	muon_IdISO.txt	Yes
Reco SF (mu_reco)	Hard-coded table in mu_reco_sf()	Yes

The combined muon weight is applied to form weight_factor:

```
weight_factor = weight_sample * mu_trig * mu_idiso * mu_reco
```

Pixel radiation damage weight (`sinal.py`)

Applied **after** the track fiducial cut (step 8 in the cutflow).
Corrects for pixel detector efficiency loss due to radiation damage.

$$w_{\mathrm{rad}} = \varepsilon_{\mathrm{multi,arm0}} \times \varepsilon_{\mathrm{rad,arm0}} \times \varepsilon_{\mathrm{multi,arm1}} \times \varepsilon_{\mathrm{rad,arm1}}$$

Each ε is read from a 2D histogram binned in track (x, y) position.

Files: - `pixelEfficiencies_multiRP_reMiniAOD.root` (multi-RP reconstruction efficiency) - `pixelEfficiencies_radiation_reMiniAOD.root` (radiation damage efficiency)

Era assignment (by event index fraction of total):

Fraction	Era	Histogram key
≤ 0.21	2018A	`h56_220_2018A_all_2D`, `h45_220_2018A_all_2D`
0.21–0.29	2018B1	`h56_220_2018B1_all_2D`, `h45_220_2018B1_all_2D`
0.29–0.37	2018B2	`h56_220_2018B2_all_2D`, `h45_220_2018B2_all_2D`

Final weight stored in output tree

The value written to the `weight` branch (used by all downstream code):

$$w_{\text{final}} = w_{\text{norm}} \times w_{\text{muon}} \times w_{\text{rad}}$$
$$= \left(w_{\text{SM}}[0] \cdot \frac{L\sigma_{\text{SM}}}{\sum w_{\text{SM}}} \right) \times (\text{SF}_{\text{trig}} \cdot \text{SF}_{\text{ID+ISO}} \cdot \text{SF}_{\text{reco}}) \times w_{\text{rad}}$$

Code location: `sinal.py`, line 966:

```
scalars["weight"][0] = weight_factor * weight
```

Background normalization (DY, ttbar)

The `weight` branch written by the background processing code contains:

$$w_{\text{DY}} = w_{\text{gen}} \times \text{SF}_{\text{muon}} \times P_{\text{proton}}$$

where $P_{\text{proton}} = 0.245$ is the proton acceptance (fraction of MC events with pileup protons on both arms after mixing).

This weight is then scaled in the plotting/shape code by:

$$\text{DY scale} = \frac{L \times \sigma_{\text{DY}}}{\sum w_{\text{gen}}} = 1.004 \times 10^{-4}$$

Sample	σ	$\sum w_{\text{gen}}$	Scale applied
DY	6077.22 pb	3.323×10^{12}	1.004×10^{-4}
ttbar	831.76 pb	(via 0.15 empirical factor)	1.0
QCD	–	–	1.0 (data-driven)
Data	–	–	1.0

Summary table

Sample	w_{norm}	$w_{\text{muon SFs}}$	$w_{\text{rad damage}}$	Proton acceptance
Signal (SM)	$w_{\text{SM}}[0] \cdot L\sigma/\Sigma w$	Yes	Yes	Explicit per-arm cut
DY	$w_{\text{gen}} \times 1.004 \times 10^{-4}$	Yes (in branch)	No	Mixed in ($P = 0.245$)
ttbar	$w_{\text{gen}} \times 1.0$	Yes (in branch)	No	Mixed in ($P = 0.245$)
QCD	1	No	No	Real protons required
Data	1	No	No	Real protons required